Project: AI-Assisted Surgical Time Risk Classification and Scheduling Support

Meeting #1: Reverse Pitching

Objective

To support surgical scheduling decisions by providing a clear, risk-based assessment of expected procedure duration relative to standard estimates

Background

Surgical schedules are commonly planned using average procedure times. However, actual surgery duration can vary significantly depending on patient-specific factors, procedural complexity, surgeon experience, and operating room conditions. This mismatch often leads to overruns, idle time, or last-minute cancellations.

Rather than producing a single time estimate alone, there is value in providing **a simple, interpretable risk signal** that communicates how likely a case is to deviate from expectations.

Proposed Output: RAG Risk Indicator

The AI system produces a **Red / Amber / Green (RAG)** classification for each scheduled surgery, based on predicted deviation from expected duration.

- **Green – Expected Duration**
  - Surgery duration is likely to fall within the planned time window.
  - No significant risk factors identified.
  - Standard scheduling assumptions are appropriate.

- **Amber – Moderate Risk of Overrun or Underrun**
  - Some patient or procedural factors suggest possible deviation from average duration.
  - Scheduling buffers may be considered.
  - Increased awareness recommended for theatre planning.

- **Red – High Risk of Significant Deviation**
  - Strong indicators that surgery may take substantially longer (or shorter) than expected.
  - Additional buffer time or schedule adjustment recommended.
  - Allows proactive mitigation of delays and downstream disruption.

System Inputs

The risk classification is generated by analysing a combination of:

- Patient medical history and co-morbidities
- Planned surgical procedure details
- Historical data from similar cases
- Surgeon-specific and theatre-specific performance patterns
- Contextual factors (e.g., time of day, case order)

Decision Logic

Rather than replacing clinical judgement, the system:

- Estimates a distribution of possible surgery durations
- Compares this distribution to the planned schedule window
- Assigns a RAG status based on the likelihood of meaningful deviation

This allows scheduling teams to interpret risk at a glance without needing to understand the underlying model.

Expected Benefits

- Clear, interpretable scheduling support for non-clinical staff
- Reduced theatre overruns and underutilisation
- Improved alignment between surgeons, schedulers, and theatre teams
- Better anticipation of downstream delay.

Extension to Postoperative Planning

The same RAG framework can later be applied to postoperative recovery planning, where:

- **Green** indicates expected recovery trajectory
  Amber indicates potential delays or increased monitoring needs
- **Red** indicates high risk of complications or prolonged recovery

Reverse Pitch Script:

- Currently, we schedule theatres based on “average” duration of procedure, (sometimes we might account for “more surgically difficult” patients with longer operation times)
- The issue with this approach, is that we either can underestimate or even overestimate the times surgical lists take, this is can lead to
    - Late Finishes
    - Last-minute Cancellations
    - Empty Theatres
- This project aims to develop a solution that more accurately estimates Surgical Procedural Times using Patient-specific Variable, ultimately, reducing uncertainty, improving surgical scheduling & improving patient outcomes

Risk Management:

- We hope our AI system will analyse multiple inputs including
    1. Patient specific factors such as age, co-morbidities, previous surgeries and their complications from patient’s pre-op assessment
    2. Procedure details including type, expected difficulty and historical data from similar cases
    3. Contextual factors such as time of day, case order and staffing

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Inputs:
- Patient medical history and co-morbidities -
- Surgical procedure details -
- Historical data from similar surgery cases -
- Surgeon and theatre specific performance patterns -
- Other data that could influence, time of day, or temperature -

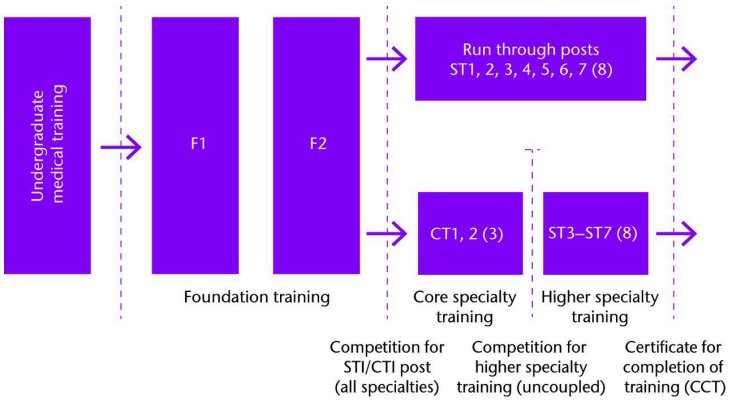
<u>The AI or Machine Learning side of things:</u>
Decision Logic:
- Layer 1: Prediction engine - for predicting durations
- Layer 2: Large Language model - incorporating structured features, and model explanations to deliver a human-friendly output and rationale

Outputs:
- 🟢 **Green:** P80 ≤ booked = low risk of overrun
- 🟠 **Amber:** P50 ≤ booked < P80 = moderate risk
- 🔴 **Red:** booked < P50 (or P90 much higher) = high risk of overrun

#Meeting 2:

<u>How to decide what initial data to use?</u>
- Public datasets
- Which procedures
- Patient side and clinical staff side factors
- Mock up data

Level of Experience → Either put no. of Cases an individual performed OR T



Booking in for Surgery

Initial Data:

- Surgeon-Related Factors (Training/Performance Metrics) = Average Procedural Time
- Level of Surgeon: **CT1, CT2 (Core Trainees)**, **ST1-3 (Junior Registrars)**, **ST3-5 (Intermediate Registrar)**, **ST5-7 (Senior Registrars)**, **Consultant (< 5 years)**, **Consultant (> 5 years)**
    - Number of Cases → People who specialise in a certain procedure will be more likely to accurately predict how long a surgical procedure would take, So model/software should target those surgeons who are more likely to underestimate or overestimate time it takes
        - Maybe helps Experienced surgeon to estimate more accurately how long their procedures take and allows more detailed analysis of their surgical performance
    - Specific Type of Surgery
    - Type of Surgery
        - Keyhole / Laparoscopic
        - Open
        - Robotic (Keyhole but controlled by the Davinci Robot) - to factor in equipment setup time
    - Personality/Style Type Scale (e.g. Fast/Efficient or Risk Adverse) → 0-10
        - Own Surgeon Profile → Can be Fixed e.g. Risk Adverse Person → Prefers to be Risk Adverse
    - Data Inputted
        - Initial Surgeon who Books the Surgery
        - Surgeon on the Day
        - Patient Related Factors
- **Patient-Related Factors - 0-10 Weighing Scale**
    - Baseline expectation of task duration (Final Say → Either Compare to the Model Prediction OR Incorporate Model Prediction) → How long the procedure the surgeon estimates it to be?
        - Fun metric → How long was I off to the actual procedural time?
        - 3 main results
            - Our Model Prediction
            - Surgeon's Prediction
            - (Actual Time the Procedure Takes) → Feedback loop
    - Intraoperative Surgical Complications
        - High-Risk of Intraoperative Complications → Weighting Scale
        - Medium-Risk of Intraoperative Complications → Weighting Scale
        - Low-Risk of Intraoperative Complications → Weighting Scale
    - (Anaesthetic Complications)* → This we can avoid for the time being
    - Past Medical History e.g.
    - Past Surgical History e.g. Cholecystectomy

Weightings for Estimations:

- Previous Cholescystectomy x 1.5 of Average Procedural Time
- Will need Power of High No. of Samples/Patients/Data
- Statistical Tests required
- Statistical Modelling can account for multiple variables through randomisation
- Even with lots of data e.g. 20 years, surgical procedures have advanced over time, so data previously might not be applicable

User Interface (What does it look like and how does it fit into Surgeon Workflow?):

- Past medical history inputting: provide categorised options for Surgeon to save time?
- Notification System e.g. Text
- Connected to EHR

Consequences:

- Flexibility & adaptability
- Easing the pre-operational assessment and decision making

MoSCoW Priorisation:

- Must Features
- Should have Features
- Could Have Features
- Won't Have Features

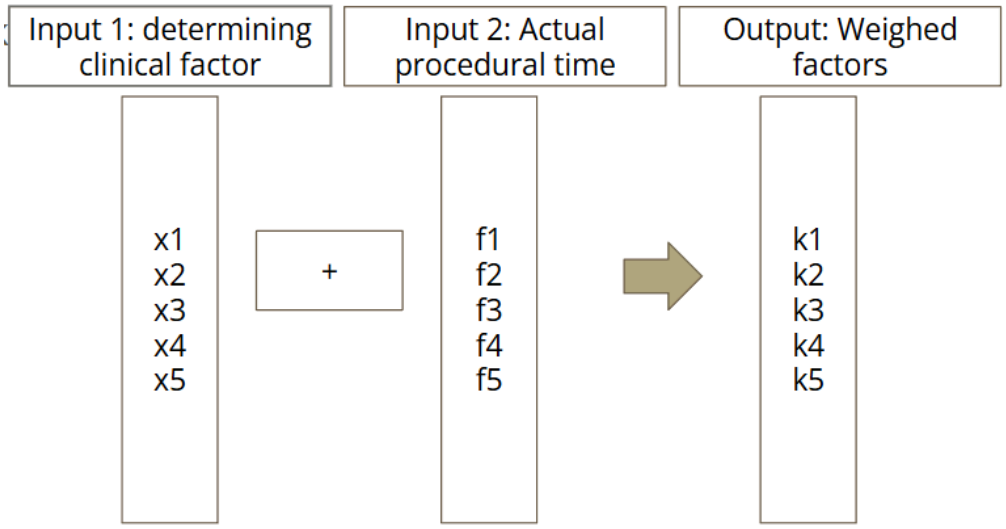
Action Plan:

- Decide what initial data we will use?
- From initial data create mock data sets for patients
- Deciding Mathematical Equation for How Long Procedure Will Take considering
- Define main user's concern - how does the mis-estimation affect the surgeons and surgeon trainees
- Research for what is the most important factors that affect the actual procedural time
- Look at regulations on medical software → GDPR, Data Protection Laws, How we are going to get the Data?

Possible features:

Must have:
Estimated procedural time as an output

Should have:
Math model - two steps: 1. Define equation - weigh the various factor by inputting the datasets that had a defined actual procedural time and the contributing factors



2. Using that equation to calculate the estimated time

Estimated time = k1x1+k2x2+k3x3…
Could have:
Complication prediction

Just ideas:

- Using Feedback Loops for Data
- Using 1 year of previous data on a specific surgery by e.g. CT1 & CT2 → To Calculate Weighted Factor
- Weighted Factor (only forms one small slice/percentage of piechart) in context of multiple different weighted factors that contribute to it
- Choose the most highly weighed factor from the trained model - piechart of factors generated by the trained model then only the highest weighed factors are brought forward

Generation of graphic is something that can be made used of?

- Generate 1st POV of surgeons from real surgical recordings for surgeons trainee's reference purpose
- Visualisation of possible complication / anatomy

Feature ideas

	User requirement	Design goal	Design feature
Idea 1: Procedural time prediction			
1.	Surgical trainees need more support from more experienced staff. The staff need a more predictable schedule to provide support for trainees.	Predict procedural time accurately. Real time update of prediction time from the operation theatre	Math model to take in factors that affect procedural time.
2.	Overrunning in a tight schedule is emotional draining.	Predict procedural time accurately.	Math model to take in factors that affect procedural time.
3.	Unexpected complication is a main reason of overruning because it is not obvious from the patient database. Surgical admission staff need more information or assistance in identifying possible complication.	Complication prediction to be provided to surgical admission staff	AI identifies hidden concerns that might cause complication. Patient data is directly pulled from EHR to train the model to identify what is the most important factors
4.	Surgical Trainees often underestimate or overestimate procedural times, there is no performance metric to assess to accurately assess whether they do this. If a metric associated with a software can show gains overtime this can prove useful for Surgical Trainees		
Idea 2: generate 1st POV of surgeons from real surgical recordings for surgeons trainee's reference purpose			
5.	Surgeon trainee's are lack of scrubbing in opportunities or cannot get a thorough view of the surgical procedure due to the fast pace in the operation theatre.	The design gives a more realistic view of the surgical procedure as in what they would need to do if they are participating in the procedure. Help trainees achieving faster progress	By recording the real procedures from different angles, the AI generate the POV of the surgeons

Concept Selection Matrix

Criteria	Possible Features			
Input data	EHR patient database	CT1&CT2 surgical data	Actual procedural duration	Procedural type which dictates any prep / equipment set-up time
Programme features a	AI to be trained from actual patients and surgeons information and duration to calculate weighted factors for each clinical and operators factors	AI to predict complication from hidden causes in EHR patient database	AI to record real cases from different angles and generate a video from surgeon's POV for trainees' revision or predicted complication visualisation	Feedback loop to compare staff/trainees- and programmed- predicted time and actual procedural time / duration taken in procedural simulation training sessions
Feedback to surgeons	RAG system	Most prevalent causes of delay displayed clearly	AI-generated predicted complication visualisation	
Feedback to trainees	Suggestion to improve their accuracy in predicting procedural time e.g. what considerations the trainees omitted?	Suggestion to improve procedural time e.g. what could they do to shorten procedural time?	AI-generated predicted complication visualisation	

User requirements:

Background

From a clinical research fellow who had general surgery experience as a FY2 doctor:

The most difficult part of surgical training:

- The demand from the surgical admissions unit is too high (i.e. too many cases)
- The hours are too long. The trainee has to make decisions after decisions (edit: exhausting and lowering efficiency)
- Not enough support. Sometimes the nurses and other doctors aren't helping and she felt like she was left alone
- Surgical training also involve seeing patient after A&E to decide whether to admit to surgery or not, apart from simulation and scrubbing in

Impact of under-estimation of procedural time:

- (doesn't happen too often) When it happens, the subsequent cases will also be delayed or have to be cancelled or be performed by another surgeon
- At young trainee level it is not really an issue, because it's the surgeon in charge (the consultant) who will make the decision (edit: the main stakeholder will be the consultant)
- If it is the last case of the day, the surgeon will just have to finish late
- It can be emotionally draining (edit: especially with high demand from the surgical admission unit, they most like already had back-to-back schedule)
- affect everyone differently

The most prominent factors that causes overrun:

- Unexpected complicated anatomy or complication during surgery. It is not obvious from the patient database that it causes delay. Normally if there is expected complexity, they would factor in more time for the surgery (edit: hidden concern)
- Patient not ready (no porters or they ate breakfast when they need to be fasted)

Medical Device Regulations for Software

- Not classed as a medical device
- Intended purpose: decisions support - it only provide reference information and the clinical decision is done by healthcare professionals with their knowledge
- If used with hardware, MDR is considered for the system as a whole. No validation explicitly for the software itself
- But if the software is used alone and directly impacting the treatment, validation has to be performed on the software
- Even if the device has to be regulated, verification and validation activities for our design is feasible
- Verification: confirm confident level of getting correct results with mock-up data beforehand
- Validation: clinical trial and feedback through interview once adopted in clinical settings

Meeting 2 notes
Input options should be more straightforward to the users - more clear definition
Complexity - how hard the surgeon deem to be
Training parameter come be turned into a game to get a better idea of how to perform the surgery in a more efficient way
How to better fit in the NHS way of working?
Workflow: surgeon decide which treatment is most suitable, and the potential risk for patients

Possible idea: doctor profiles - experience (no. of procedure - depends on training pathway and core competency: show the competency to progress), what type of doctor they are
eLogbook - for progression purpose, does not have the matrix of how long was the surgery
Possible need for training matrix

Surgeon's prediction
Model's prediction
Actual time used
- To be feedback loop for training for time prediction

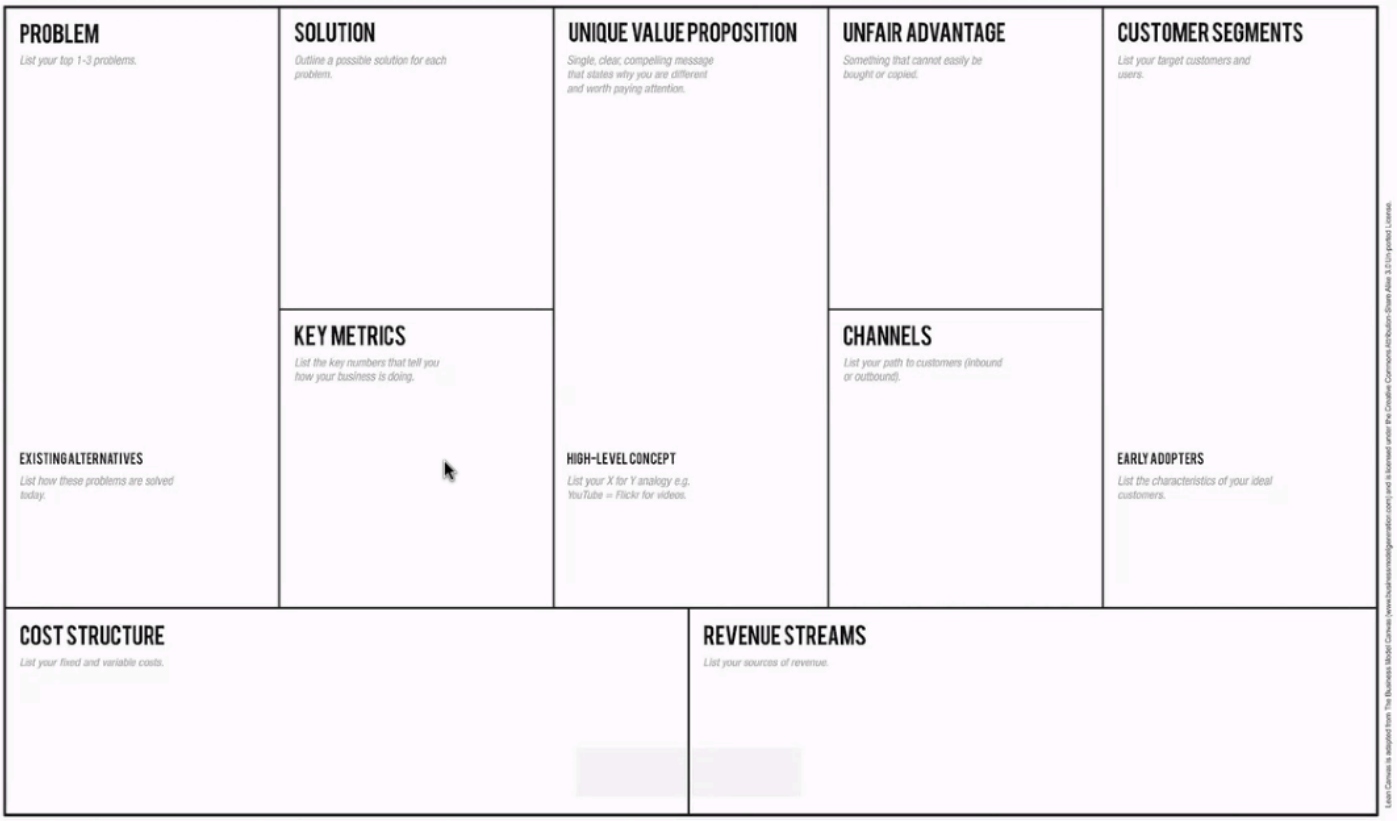
To focus on the surgeon-related factor - trainees often underestimate the time
Estimation of time can be really research-heavy, might be better to just focus on the surgeon-related factor
Life-long journey - easy to use, not add to the schedule

How to pitch about improving patient outcome

Questions:
- Why has no one built it before?
- Improves prediction from 50% to 90%?
- Machine Learning Engineers input?
- Pick a specific procedure initially?
- How did you show it was 10x value?
- Likert Scale/MS Scale/Net Promotional Value → How likely are you to refer to us to another doctor?
- Create Mock Data
- Attach Emotional Data

Your Tasks:

1. Speak to customers
2. What is your value proposition?
3. Who pays, who uses, who sabotages?
4. What price are they willing to pay? How often?
5. What are your costs to get to MVP?



National Joint Registry:

- https://surgeonprofile.njrcentre.org.uk/SurgeonProfile?gmccode=4552202
- Ethics About Data → Sharing Profile